Seongmoon **Jeong**

PHD CANDIDATE

IRIS Lab., Sungkyunkwan University, Suwon, Republic of Korea

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I am a Ph.D. candidate in the Department of Artificial Intelligence at Sungkyunkwan University, advised by **Prof. Jong Hwan Ko**. My research focuses on **image compression for machine perception**, spanning low-level optimization and high-level tasks such as classification, object detection, and segmentation. I develop **learned, task-aware representations** that jointly optimize compression efficiency and downstream vision performance. More recently, I have investigated high-level vision directly in the **RAW domain**, before the image signal processing (ISP) pipeline, to preserve information relevant to machine perception. My long-term goal is to enable **communication-efficient intelligent systems** for autonomous systems and edge-based machine vision. **Available for full-time roles starting March 2027.**

Education

Ph.D. in Artificial Intelligence (Expected February 2027) — *Suwon, Korea*

SUNGKYUNKWAN UNIVERSITY (SKKU) — *Mar. 2020 - Present*

- Received an Outstanding Scholarship awarded to promising students in the Department of Artificial Intelligence.

B.S. in Electronic and Electrical Engineering — *Suwon, Korea*

SUNGKYUNKWAN UNIVERSITY (SKKU) — *Mar. 2013 - Feb. 2020*

- GPA: 3.84/4.5 (Top 6.6%)

Research Experience

Graduate Researcher — *Suwon, Korea*

IRIS LAB, SUNGKYUNKWAN UNIVERSITY (SKKU) — *Mar. 2020 - Present*

- General Codec Adapter for Machines
- Learned Image Compression Adapter for Machines
- Efficient JPEG Modification for Machines

Project Experience

Video Coding for Machines — *Media Research Division, ETRI*

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU) — *2023 - 2024*

Adaptive and Efficient Preprocessing and Coding for Machine Vision Tasks and Input Images — *Media Research Division, ETRI*

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU) — *2022*

Selected Publications

Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection — *IEEE AVSS*

1ST AUTHOR — *Sep. 2026*

UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression — *IEEE AVSS*

1ST AUTHOR — *Sep. 2026*

Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability — *IEEE CVPR*

2ND AUTHOR — *Mar. 2025*

Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression — *IEEE AICAS*

2ND AUTHOR — *Feb. 2024*

Rate-Controllable and Target-Dependent JPEG-Based Image Compression Using Feature Modulation *IEEE ICME Workshop*
1ST AUTHOR *Apr. 2023*

An Overhead-Free Region-Based JPEG Framework for Task-Driven Image Compression *Pattern Recognition Letters*
2ND AUTHOR *Nov. 2022*

Standardization Activities

[VCM] Neural network based post-filter for VCM *Geneva*
ISO/IEC JTC1 / SC29 / WG4 M71305 *Jan. 2025*

[VCM] Neural network based pre-filter for VCM *Geneva*
ISO/IEC JTC1 / SC29 / WG4 M71304 *Jan. 2025*

[VCM] Learned joint filter network for VCM *Kemer*
ISO/IEC JTC1 / SC29 / WG4 M69990 *Nov. 2024*

[VCM] Learned pre-filter network for VCM *Sapporo*
ISO/IEC JTC1 / SC29 / WG4 M69085 *Jan. 2024*

Honors & Awards

2024	**3rd Place**, DAC System Design Contest - GPU Track	*San Francisco, USA*
2020	**2nd Place**, Artificial Intelligence Grand Challenge, Ministry of Science and ICT	*Korea*
2019	**Honorable Mention**, Big Data Center Open Innovation Challenge	*Seongnam, Korea*
2019	**Outstanding Work Award**, Capstone Design Project, SKKU	*Suwon, Korea*
2018	**Dean's List Award**, Fall semester, SKKU	*Suwon, Korea*
2017	**Dean's List Award**, Fall semester, SKKU	*Suwon, Korea*

Full Publications

INTERNATIONAL CONFERENCE

Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection *IEEE AVSS*
1ST AUTHOR *Sep. 2026*

UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression *IEEE AVSS*
1ST AUTHOR *Sep. 2026*

Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability *IEEE CVPR*
2ND AUTHOR *Mar. 2025*

INTERNATIONAL CONFERENCE (CONTINUED)

Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression — *IEEE AICAS*
2ND AUTHOR — *Feb. 2024*

Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays — *IEEE ICCAD*
4TH AUTHOR — *Jul. 2023*

Rate-Controllable and Target-Dependent JPEG-Based Image Compression Using Feature Modulation — *IEEE ICME Workshop*
1ST AUTHOR — *Apr. 2023*

INTERNATIONAL JOURNAL

A Codec-Transferable Adapter for Machine-Oriented Image Compression Across Multiple Vision Tasks — *Preprint (Under review for TCSVT)*
1ST AUTHOR — *2025*

KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping — *IEEE TCAS-I*
4TH AUTHOR — *Feb. 2024*

An Overhead-Free Region-Based JPEG Framework for Task-Driven Image Compression — *Pattern Recognition Letters*
2ND AUTHOR — *Nov. 2022*

Student Mentoring

Hangyul Choi
M.S. STUDENT AT SUNGKYUNKWAN UNIVERSITY, NOW AT MX BUSINESS, SAMSUNG ELECTRONICS — *2023 - 2024*

- Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression

Chanung Park
M.S. STUDENT AT SUNGKYUNKWAN UNIVERSITY — *2024*

Skills

Programming Languages	Python, Shell scripting, C/C++
ML Frameworks	PyTorch, TensorFlow, JAX, TensorRT
Development Frameworks	Docker, Conda, uv

Reference

Jong Hwan Ko — *Ph.D. advisor*
ASSOCIATE PROFESSOR — *Dept. of ECE, SKKU, Suwon, Korea*

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- jhko@skku.edu